

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-012: Dark Mode / Light Mode Theme Toggle

University College of Jaffna
Department of Interdisciplinary Studies
Information and Communication Technology
SCM (K72C001M12)
NVQ Level 6 – Semester 1

Change Request Details

Field	Details
Change Request ID	CR-012
Change Request Title	Dark Mode / Light Mode Theme Toggle
Student Name	S.Hindusha
Student ID	JFICT2426
Implementation Status	Implemented – Testing Required
Date	9 October 2026

1. Change Request Summary

1.1 Business Requirement

The Student Management System originally used a light theme across its user interface. Users required an option to switch between light mode and dark mode to improve visual comfort, readability, and user experience.

The change was intended to provide theme switching functionality while preserving the existing application features, page layouts, authentication process, and student management operations.

1.2 Existing Behaviour

Before implementation:

- The application primarily displayed a light-coloured interface.
- Page backgrounds, headers, cards, forms, and tables used fixed CSS colours.
- Users could not independently select their preferred display theme.
- The existing login, administrator dashboard, and student dashboard functionality was already available.
- The application used React components and separate CSS files for individual pages.

1.3 Requested Change

The following functionality was required:

1. Provide a dark mode and light mode toggle.
2. Add the theme toggle to the appropriate page header.
3. Apply the selected theme to the complete relevant page, rather than changing only the toggle button.
4. Support the administrator dashboard and student dashboard.
5. Preserve existing dashboard content, including profile information, buttons, tables, forms, and navigation.
6. Maintain appropriate text contrast and readability in both themes.
7. Save theme preferences so that the selected theme can be restored.
8. Avoid changing backend functionality or database records solely to support theme switching.

2. Implementation

2.1 Implementation Summary

The theme functionality was designed using React state, a shared theme context, and CSS custom properties.

The theme context manages the selected theme for the relevant application section. The selected theme is applied to the document using the data-theme attribute.

The CSS styles use variables for page backgrounds, surfaces, text, and borders. This allows the appearance of the interface to change when the selected theme changes.

The implementation must ensure that all relevant elements use the theme variables. Updating only the toggle button or a small number of components is insufficient to apply dark mode throughout the page.

2.2 Affected Files

File	Change	Purpose
frontend/src/context/ThemeContext.jsx	Modified / Added	Manages theme state, section preferences, and document theme application.
frontend/src/components/ThemeToggle.jsx	Modified / Added	Provides the theme toggle control.
frontend/src/pages/DashboardPage.jsx	Modified	Integrates the theme toggle into the administrator dashboard header.
frontend/src/pages/StudentDashboardPage.jsx	Modified	Integrates the theme toggle into the student dashboard header.
frontend/src/styles/global.css	Modified	Defines global theme variables and shared theme styles.

frontend/src/styles/dashboard.css	Modified	Applies theme variables to administrator dashboard components.
frontend/src/styles/studentDashboard.css	Modified	Applies theme variables to student dashboard components.
frontend/src/App.jsx	Modified if required	Wraps the application with the theme provider.

2.3 Technical Changes

Theme State Management

React state is used to track the selected theme.

The supported theme values are:

- light
- dark

The initial theme can be restored from browser local storage. If no valid preference exists, the application can use light mode as its default.

Theme Context

A shared React context provides theme information and theme-switching functions to the relevant components.

The context can manage separate preferences for the login page, administrator dashboard, and student dashboard when independent themes are required.

Theme Toggle

The toggle component allows the user to change the selected theme.

The selected theme should determine the appearance of the entire relevant page. The toggle should not interfere with logout, navigation, login submission, or other existing buttons.

CSS Custom Properties

CSS variables are used to support both themes.

Example:

```
:root,  
html[data-theme="light"] {  
  --page-bg: #f8fafc;  
  --surface-bg: #ffffff;  
  --text-primary: #0f172a;  
  --text-secondary: #64748b;  
  --border-color: #e2e8f0;  
}
```

```
html[data-theme="dark"] {  
  --page-bg: #0f172a;  
  --surface-bg: #1e293b;  
  --text-primary: #f8fafc;  
  --text-secondary: #cbd5e1;  
  --border-color: #334155;  
}
```

Dashboard backgrounds, headers, cards, form fields, tables, and text elements must reference these variables for the theme to apply consistently.

Existing Functionality Preservation

The theme change should preserve:

- Existing administrator dashboard functionality.
- Existing student profile information and dashboard layout.
- Login and logout behaviour.
- Student creation, editing, and deletion.
- Existing API integration and validation.
- Existing routes and role-based access control.

The theme functionality is primarily a frontend change and does not require a backend endpoint or database modification.

2.4 Implementation Logic

1. Load the application and initialize the theme preferences.
2. Restore a valid saved preference when available.
3. Display the relevant page using the selected theme.
4. Display the theme toggle in the appropriate header.
5. When the user activates the toggle, update the theme state.
6. Apply the selected theme to the document.
7. Update all CSS elements that use the theme variables.
8. Save the preference in local storage when appropriate.
9. Preserve the current page content and functionality while changing the appearance.

2.5 Error and Data Handling

The implementation should validate the saved theme value and fall back to light mode if the stored value is invalid or unavailable.

If browser storage is unavailable, the application should still function using its default theme.

Changing the theme must not clear user profile data, modify student records, or trigger unnecessary API requests.

3. Testing & Verification

3.1 Testing Approach

The implementation should be verified through manual frontend testing.

Testing types:

- Functional testing
- User interface testing
- Theme persistence testing
- Accessibility testing
- Regression testing
- Responsive design testing
- Role-based dashboard testing

Environment: Development

Browser: Microsoft Edge / Google Chrome

Test roles: Administrator and Student

3.2 Test Cases

Test ID	Test Case	Expected Result
TC-01	Open the application in light mode.	All relevant components display the light theme.
TC-02	Switch to dark mode.	The entire relevant page changes to dark mode.
TC-03	Switch back to light mode.	The page returns to light mode.
TC-04	Inspect the administrator header and content.	Header, cards, tables, and text use the selected theme.

TC-05	Inspect the student dashboard.	Profile card, details, header, and background use the selected theme.
TC-06	Refresh the page after selecting a theme.	The saved theme preference is restored where configured.
TC-07	Log in as an administrator.	The administrator dashboard and its existing controls remain available.
TC-08	Log in as a student.	The student dashboard and profile information remain available.
TC-09	Test buttons, forms, and tables in both themes.	Existing functions work and remain readable.
TC-10	Test the application at a narrow screen width.	The layout remains usable and responsive.
TC-11	Test with no saved theme preference.	The application uses the configured default theme.
TC-12	Check browser console and network activity.	No new errors or unnecessary requests are introduced by theme switching.

3.3 Test Summary

Test Result	Count
Total Test Cases	12
Passed	To be completed after testing
Failed	To be completed after testing
Pass Rate	To be calculated after testing

Overall Test Result: Pending verification.

4. Acceptance Criteria Verification

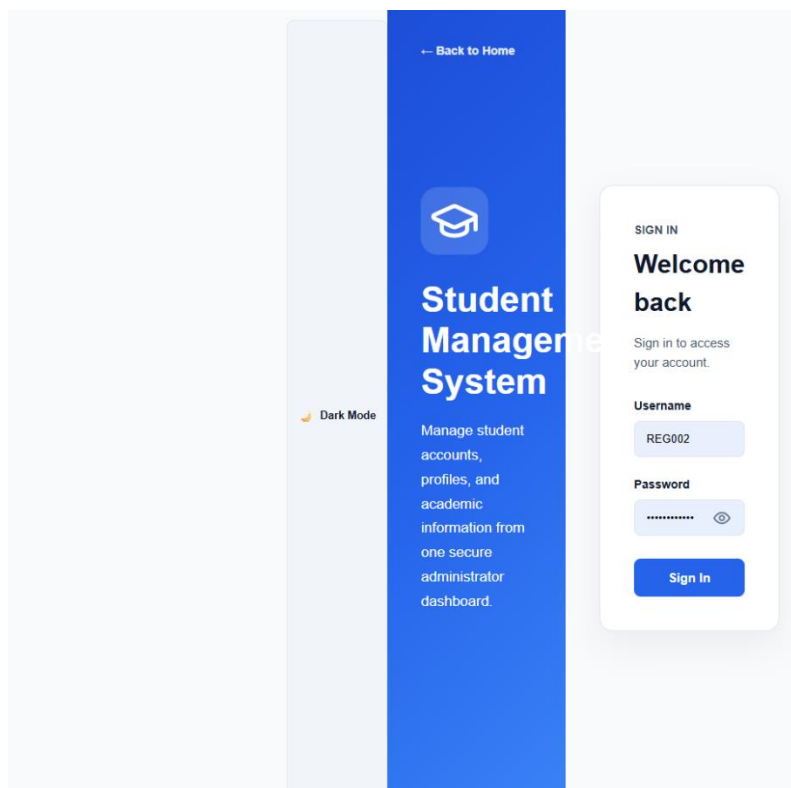
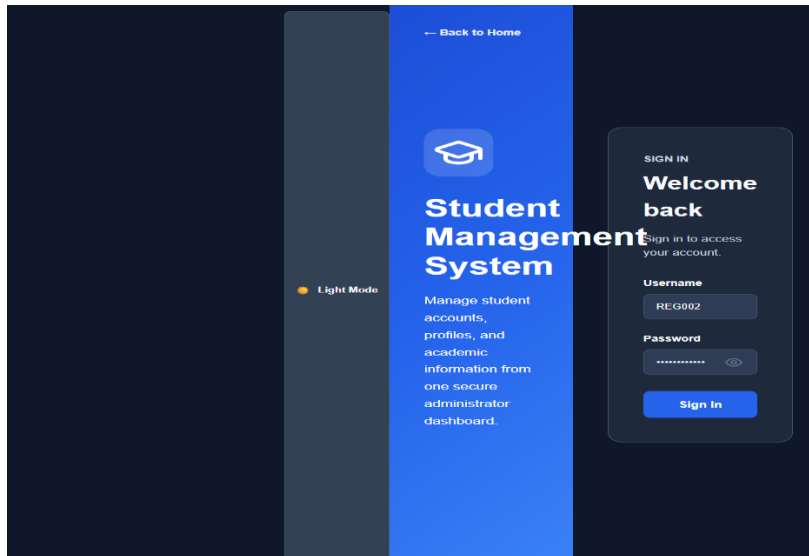
ID	Acceptance Criterion	Verification Method	Status
AC-001	Users can switch between light and dark mode.	Theme toggle test	Pending verification
AC-002	The theme applies to the entire relevant page.	Visual inspection	Pending verification
AC-003	The administrator dashboard remains functional.	Regression testing	Pending verification
AC-004	The student dashboard remains functional.	Regression testing	Pending verification
AC-005	Text remains readable in both themes.	UI and contrast inspection	Pending verification
AC-006	The theme preference is restored when configured.	Refresh test	Pending verification
AC-007	Existing login and logout functionality is preserved.	Functional testing	Pending verification
AC-008	Existing student management operations remain available.	Regression testing	Pending verification
AC-009	The responsive layout is maintained.	Responsive UI testing	Pending verification
AC-010	Theme switching does not require backend changes.	Implementation review	Pending verification

5. Evidence

A maximum of two relevant screenshots should be attached.

Evidence 01 – Administrator Dashboard Theme

Related: AC-001, AC-002, AC-003, AC-005

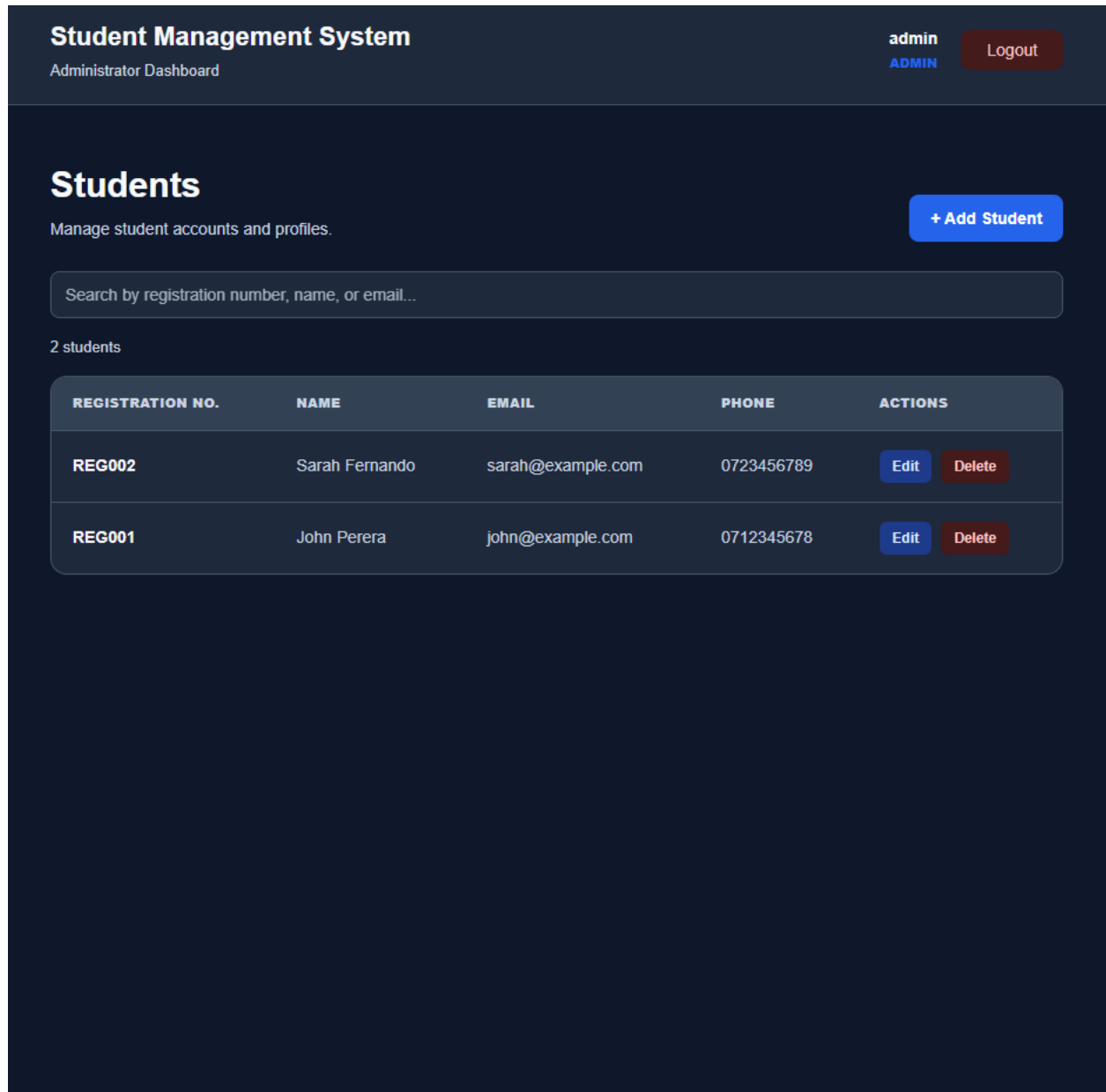


Insert a screenshot of the administrator dashboard in dark mode, showing the header, theme toggle, student search, Add Student button, and student table.

Figure 1: Administrator dashboard displaying dark mode.

Evidence 02 – Student Dashboard Theme

Related: AC-001, AC-002, AC-004, AC-005, AC-006



Students

Manage student accounts and profiles.

+ Add Student

Search by registration number, name, or email...

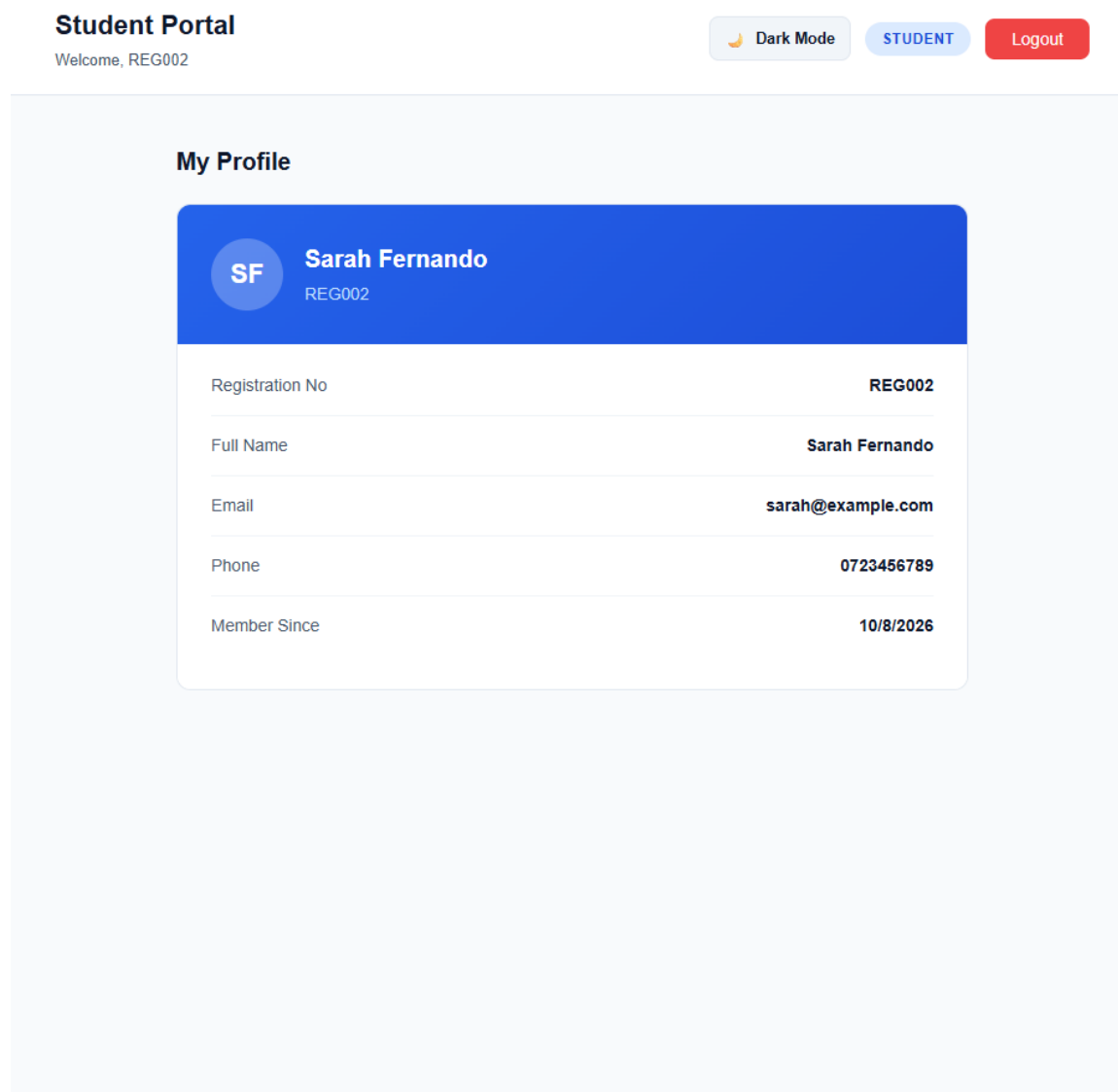
2 students

REGISTRATION NO.	NAME	EMAIL	PHONE	ACTIONS	
REG002	Sarah Fernando	sarah@example.com	0723456789	Edit	Delete
REG001	John Perera	john@example.com	0712345678	Edit	Delete

Insert a screenshot of the student dashboard in dark mode, showing the header, theme toggle, profile card, and profile details.

Figure 2: Student dashboard displaying dark mode.

Privacy note: Avoid including passwords, authentication tokens, or unnecessary sensitive student information in screenshots.





My Profile



Sarah Fernando

REG002

Registration No	REG002
Full Name	Sarah Fernando
Email	sarah@example.com
Phone	0723456789
Member Since	10/8/2026

6. Final Result

Item	Result
Requirements	To be verified
Implementation	Theme context and toggle integration implemented or under development
Testing	Pending verification
Acceptance Criteria	Pending verification
Existing Dashboard Functionality	Must be confirmed through regression testing
Backend Changes	Not required for theme switching
Final Status	Pending testing and evidence review

6.1 Final Summary

CR-012 introduces dark mode and light mode functionality to the Student Management System. The feature uses React state, a shared theme context, browser local storage, and CSS custom properties to manage and apply the selected theme.

The intended implementation supports the administrator and student dashboards while preserving existing login, navigation, profile display, and student management functionality.

The theme must be applied consistently across page backgrounds, headers, cards, tables, forms, and text. The implementation should be accepted only after confirming that both dashboards remain complete and functional in both themes.

6.2 Issues / Limitations

The theme will not change the entire interface if individual CSS rules continue to use fixed light-mode colours. All relevant components must use the appropriate theme variables.

Theme persistence depends on browser storage availability. The selected theme and the active dashboard section must also remain synchronized when users navigate between pages.

No backend or database changes are required solely for frontend theme switching.

Student Name: S.Hindusha

Student ID: JFICT2426

Date: 9 October 2026

Signature: _____

END OF REPORT